Project Proposal

AirWatch India: Multi-City Air Quality Analysis and Interactive Dashboard

1. Project Title

AirWatch India: Multi-City Air Quality Analysis and Interactive Dashboard

2. Introduction

Air pollution is one of the most serious public health and environmental challenges in urban India. Citizens, researchers and policymakers need clear, accessible information about how polluted the air is, which pollutants drive it, and how conditions change over time. This project analyzes ten years of daily air quality data for five major cities and presents the findings through an interactive dashboard.

3. Problem Statement

Air quality data is often scattered, hard to interpret and not presented in a way that supports quick decisions. There is a need for a single, easy-to-use tool that compares cities, tracks long-term trends, highlights the dominant pollutants and translates AQI values into understandable health guidance.

4. Objectives

- Analyze AQI and pollutant levels across Delhi, Mumbai, Chennai, Kolkata and Bangalore.
- Identify historical, seasonal and yearly trends in air quality.
- Study the relationships between pollutants (PM2.5, PM10, NO_2, CO, etc.) and the AQI.
- Compare cities and rank them by pollution severity.
- Build an interactive dashboard with charts, comparisons and health advisories.

5. Dataset Description

The project uses the cleaned dataset **cleaned_city.csv**.

- **Records:** 18,265 daily observations
- **Period:** 1 January 2015 to 31 December 2024 (3,653 days)
- **Cities:** Delhi, Mumbai, Chennai, Kolkata, Bangalore
- **Pollutant variables (12):** PM2.5, PM10, NO, NO_2, NO_x, NH_3, CO, SO_2, O_3, Benzene, Toluene, Xylene
- **Target variables:** AQI (numeric) and AQI_Bucket (Good, Satisfactory, Moderate, Poor, Very Poor, Severe)
- **Data quality:** Already cleaned; contains no missing values.

6. Proposed Methodology

The project will be developed through the following steps:

1. **Data Collection:** Obtain the air quality dataset for the five cities and, where required, supplement it with weather data (temperature, humidity, wind speed, rainfall) from reliable APIs or public sources.
2. **Data Cleaning and Preparation:** Verify data types, convert dates to datetime format, check for duplicates and outliers, and derive features such as year, month and season.
3. **Exploratory Data Analysis:** Analyze AQI distribution, pollutant levels, yearly and seasonal trends, correlations and differences between cities.

4. **Data Visualization:** Present results using line charts, bar charts, heatmaps, box plots, scatter plots and correlation matrices.
5. **Predictive Modelling (optional):** Build a model to predict AQI or AQI category from pollutant levels and evaluate it with standard metrics.
6. **Interactive Dashboard:** Develop a user-friendly dashboard with filters for city, date range and pollutant, KPIs, trends and health advisories.

7. Proposed Features

- AQI monitoring with current and average values per city.
- Pollutant monitoring for PM2.5, PM10, NO, NO_2, NO_x, NH_3, CO, SO_2, O_3, Benzene, Toluene and Xylene.
- Historical AQI trend charts (daily, monthly and yearly).
- City-wise comparison and ranking of pollution levels.
- AQI category distribution (Good to Severe) as a donut or bar chart.
- Correlation analysis between pollutants and AQI.
- Filters for city, date range and pollutant.
- Health advisories and recommended actions based on AQI category.
- (Optional) Weather integration and AQI prediction.

8. Expected Dashboard

The dashboard, **AirWatch India**, is a dark-themed command center with a filter sidebar (city, year, pollutant, category) and an AQI gauge, KPI cards (PM2.5, PM10, NO_2, share of Severe days), a city-by-year AQI heatmap, monthly AQI lines per city, a stacked AQI category mix by city, an average pollutant bar chart and a health advisory panel.



9. Tools and Technologies

- **Language:** Python
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Scikit-learn
- **Dashboard:** Streamlit, Power BI or Tableau
- **Environment:** Jupyter Notebook, GitHub for version control

10. Expected Outcomes

- A clear understanding of pollution trends and severity across the five cities.
- Identification of the pollutants that most strongly influence AQI.
- A reusable, interactive dashboard for exploring air quality.
- Actionable health guidance derived from AQI categories.

11. Timeline

- **Week 1:** Problem definition, dataset study and data preparation
- **Week 2:** Exploratory data analysis
- **Week 3:** Visualization and (optional) predictive modelling
- **Week 4:** Dashboard development and testing
- **Week 5:** Documentation, final report and presentation

12. Limitations and Future Scope

The current dataset covers five cities and contains no weather variables, so weather and AQI relationships would require additional data. Future work could add real-time API feeds, more cities, interactive pollution maps and forecasting of upcoming AQI levels.

13. Conclusion

This project turns a decade of raw air quality data into clear insights and an easy-to-use dashboard. It will help users understand pollution patterns, compare cities and make informed decisions to reduce health risks.